DS-Harness: A Rule-First Agent with an LLM Fallback

Technical Project Report & System Architecture — ML-Arena Challenge 194 · updated 7 November 2026

End of project: 2027-01-04 (end of day) | Oral: 2027-01-06, 9:00–12:00 | Private Score: 97.5 % (v10) | v5: 92.5 %

97.5 %	117 / 120	180 / 180	40 s
PRIVATE LEADERBOARD	TASKS SOLVED	DEV TASKS SOLVED BY RULES	BATCH LIMIT (8 TASKS)

1. Challenge Overview & Task Objectives

The ML-Arena Challenge 194 requires building an autonomous data science agent capable of answering diverse quantitative tasks. Each task consists of an English objective, optionally accompanied by CSV data files and a data dictionary. The required output is a single number, evaluated within a tolerance of 1.5×10^{-n} for n requested decimal places.

2. Task Taxonomy & Execution Constraints

The private evaluation set consists of 120 unseen tasks. The system runs under strict constraints: a local model of about 1.5 billion parameters (Qwen2.5-Coder-1.5B), no internet access, 3 vCPUs, 3 GiB of RAM, and 40 seconds per call of 8 tasks.

Task Family	Content & Functional Scope	Tasks	Share	v10 private
No files	Arithmetic, percentages, growth, dates, logarithms and roots, integers, probability, word problems	45	37.5 %	44 / 45
Tables	Statistics, filters, group rankings, joins, derived columns, and types absent from dev	50	41.7 %	48 / 50
Fits	Linear or logistic regression on `train.csv`	25	20.8 %	25 / 25

System Design Strategy

Deterministic rules answer every recognised template in milliseconds; on the public dev set they answer all 180 tasks without calling the model. The GPU budget is therefore kept for the tasks no rule recognises, where the LLM gets parser hints and a larger vote.

3. Overall System Architecture

DS-Harness is a two-tier design: high-precision rule solvers first, backed by an LLM code-generation pipeline with error recovery, answer checks and voting.

1. Router — No files / fit / table → **2. Ingestion** — Cleaning, units, dates, keys, thousands separators → **3. Rule engine** — Narrow templates, veto, two cross-checked rule sets → **4. LLM fallback** — Qwen code generator with parser hints → **5. Repair, checks & vote** — Error repair, answer checks, weighted vote, rounding

4. Deterministic Rule Engine (Rule-First Strategy)

The rules exploit the regularity of the task generator. A rule answers only when the whole question matches its template; otherwise it abstains and the task goes to the LLM.

- **Fits (25/25 private):** target = the `train.csv` column missing from `test.csv`; features matched to the dictionary; year filters; unregularised linear or logistic regression; predictions, intercepts, coefficients, odds ratios, R^2, counts and mean predictions.
- **Tables (48/50 private):** narrow templates for types absent from dev (share of a total, days between dates, gaps between groups, top-k, mode, interquartile range, weekday and quarter filters, date ranges, negations, `A or B`, per-group spread and distinct counts, weighted means, year-over-year change); the original v5 rules for statistics, filters, groups, joins and derived columns; a corrected rule set that cross-checks them.
- **No files (44/45 private):** the v5 templates plus about 130 further templates in three waves (dates and times, logarithms and powers, compound, simple, monthly and continuous interest, half-life, divisors, primes, binomial and hypergeometric probability, variance, mode, unit and temperature conversions, mixtures…).

Two rule sets, cross-checked (v10)

The v5 table rules are kept verbatim. A second, corrected implementation runs on the same task; when both answer and disagree, the corrected answer is kept. On generated tasks the corrected rules were right in 41 of 41 disagreements; on the private set this step added two table tasks (46 → 48).

Wrong rules are worse than abstentions

On generated tasks of types absent from dev, the v5 rules answered 50 of 420 tasks wrongly instead of abstaining (an interquartile range computed as max − min, "number of distinct carriers per depot" counted as rows, "not of the Lyon store" read as "of the Lyon store"). A wrong rule hides the task from the LLM. v10 vetoes these constructs and solves them with dedicated templates.

5. LLM Fallback & Self-Correction Pipeline

For tasks outside the rule scope, Qwen2.5-Coder-1.5B writes executable Python:

- **Compact prompting:** cleaned table schemas, dictionary, instructions and one worked example.
- **Parser hints (v9):** what the parser understood even when no rule answered — the resolved column of the measure, the grouping column, the filters, and, for proportions, the population and the counted rows kept apart.
- **SmartFrame and repair:** column-name mismatches are recovered at run time; failed code is sent back with its error message; truncated code is continued.
- **Answer checks (v8):** candidates that break the form the question asks for (a fraction or probability outside [0, 1], a non-integer count, an invalid date, a rank below 1) are discarded before the vote. None of the true answers of 9,811 tasks is rejected.
- **Weighted vote:** greedy result weight 1.5, each sample weight 1; 8 samples instead of 4 when four tasks or fewer of a batch reach the model (v9); internal budget of 29 s, every generation bounded by `max_time`.

6. Post-Mortem & Engineering Fixes

Identified Vulnerability	System Impact	Engineered Fix
LLM re-imported pandas, resetting `read_csv`	Table tasks failed silently	Patched `pd.read_csv` persistently across re-imports
scikit-learn estimators rejected missing values	Fit tasks crashed on NaN rows	NaN-tolerant regression wrappers
Inconsistent join keys (case, prefixes, leading zeros)	Join accuracy at 0 %	Automatic key detection and normalisation
Default L2 regularisation in logistic regression	Probabilities outside tolerance	Unregularised fits
Unbounded code generation	Deployment failed on deadline	`max_time` and a watched thread
Field names containing "of" (`length of stay`) cut to "stay"	Filter silently dropped	Candidate phrases tried nearest first, within the sentence
Apostrophe in "company's" read as a quote; sentinel sentence matched across sentences	Filters and `-1` sentinels lost	Word-bounded quotes, sentence-bounded patterns
`no less than 5` read as `< 5` ; synonyms (`more than` , `under` ...) ignored	Wrong filters	Explicit operator table
Thousands separators (`24,000` , `1 tonne = 1,000 kg`)	Numbers truncated	Normalised before parsing
"How much interest has been earned" answered with the balance	Wrong no-file answers	Interest-only questions handled

7. Experimental Progression & Leaderboard Results

Version	Key change	No Files	Tables	Fits	Private Score
v1	Initial agent without rule engine	75.6 %	40.0 %	32.0 %	51.7 % (62/120)
v2	Fit rules + execution recovery fixes	77.8 %	60.0 %	96.0 %	74.2 % (89/120)
v3	Expanded table and arithmetic rules	91.1 %	84.0 %	96.0 %	89.2 % (107/120)
v4	Enhanced no-file rule templates	95.6 %	84.0 %	96.0 %	90.8 % (109/120)
v5	Final rule optimisations (`stage3.py`)	95.6 %	88.0 %	96.0 %	92.5 % (111/120)
v6	v5 table path + reworked no-file and fit rules	97.8 %	86.0 %	100 %	93.3 % (112/120)
v7	+ 21 narrow table templates and veto	97.8 %	88.0 %	100 %	94.2 % (113/120)
v8	+ no-file template waves, thousands separators, answer checks	97.8 %	88.0 %	100 %	94.2 % (113/120)
v9	+ parser hints in table prompts, 8 samples	97.8 %	92.0 %	100 %	95.8 % (115/120)
v10 (best)	**+ corrected table rules win on disagreement (`agent_v10.py`)**	**97.8 %**	**96.0 %**	**100 %**	**97.5 % (117/120)**

v6–v10 were submitted on 5–6 November 2026, within the project period: scores are frozen on 4 January 2027 (end of day) at the last deployment that passed, so v10 must be the last deployment. Each version changes one thing; one private task is worth 0.83 points, and the LLM path alone moves the score by about one task between submissions.

Experiment that did not help	Private Score	Reading
Vote weighted by use of the columns found by the parser	95.8 %	No effect
12 samples and a second round on split votes	95.8 %	No effect
Hints for no-file tasks sent to the LLM	95.8 %	No effect
LLM arbitrates between the two table rule sets	96.7 %	One table recovered instead of two
Corrected rules also where the v5 rules abstain	93.3 %	Tables 86 %: generic rules misfire on private phrasings

8. Testing Methodology

The dev set was used to write the rules, so it cannot measure generalisation. Changes were therefore tested on data built for that purpose, and the LLM path, which could not be run offline, was measured only through the private score.

- **Generators with ground truth:** the seven table domains, the five fit domains and the nine no-file categories are rebuilt with dev's formatting traps (separators, decimal commas, missing-value tokens, units in cells, `-1` sentinels, duplicates, day-first dates, key prefixes). The truth is computed on the clean data, independently of the agent.
- **Paraphrases of dev:** real dev objectives rewritten without changing their meaning (operator synonyms, filler sentences, sentence order, rounding phrases, "Find" instead of "What is").
- **Strict non-regression:** every version is compared task by task with the previous one, with a stub LLM, on answers and on the exact prompts sent to the model.

Check	Result
Dev, full `solve()` with a stub LLM	180/180, no LLM call, ≤ 1.5 s per batch of 8
Generated tables (dev-like and new types, fresh draws)	1,548/1,550 correct, 0 wrong; new types 420/420 on two draws
Generated no-file tasks (three waves, thousands separators)	100 % on every set, 0 wrong
Paraphrased dev objectives	Tables 765/775, no files 542/542, fits 244/244 (v5 rules: 734, 216, 240)
Answer checks on true answers	0 rejected among 9,811 tasks

A Good-Turing estimate on dev's no-file sub-templates suggests that about a third of the private no-file tasks come from sub-templates dev does not contain, which is why the template waves cover many forms.

9. Failure Analysis & Lessons

Three of 120 private tasks are still missed: two tables and one no-file task. Their scores did not move with any change of the LLM path, so they are most likely decided by rules; task-level private errors are not available.

- **Stable scores are a diagnosis.** Five LLM-side variants gave exactly the same family scores, which pointed to deterministic errors; cross-checking two rule implementations then recovered two tasks.
- **Generated data is a safety net, not the arbiter.** Extending the corrected rules to tasks the v5 rules abstain on looked better on generated data but cost five private tasks: the LLM with hints handles unseen phrasings better than generic rules.
- **Leaderboard selection.** v6–v10 were chosen among about a dozen submissions on the same private set. Each was a single explained hypothesis, but the final score also reflects that selection.
- **Next step.** Measure the LLM path offline with the rules switched off (`DSH_RULES_OFF`) on a GPU, then target the forms where the model fails.